Alexander Aglerimov

Senior Rust Developer | Real-Time Computer Vision Backends | Autonomous Systems

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Summary

Graduated from Taganrog State Radio Technical University (Southern Federal University), B.Sc. in Computer Science and Programming, 2017. Has 10+ years of programming experience and 8+ years of commercial software development, including 5 years of commercial Rust development. Since 2023 a core backend developer of **Pulsar**, a real-time traffic-enforcement platform (computer vision, 70 Rust modules on a message bus, 500K+ LOC) deployed on roadside complexes in Russia, Belarus and Uzbekistan: delivered 100+ tasks and 800+ commits, touched 68 of 69 modules, shipped 58 production hotfixes. Earlier: 2 years of HFT / crypto-exchange connectivity (13 exchanges, WebSocket + REST) and blockchain work (Substrate, NEAR, Oasis, Solana). In 2026 independently built and open-sourced 4 R&D projects (270K+ LOC, 2,600+ tests): a GNSS-denied autonomous UAV navigation stack (VIO / LIO fused into PX4 EKF2 with map-based relocalization, benchmarked against OpenVINS on EuRoC; 3D occupancy mapping, D* Lite, GPU MPPI, deterministic replay; C++20, ROS 2, PX4 SITL, Gazebo, CUDA; 168 GitHub stars), a drone-swarm coordination runtime (Rust, MAVLink), a self-organizing agent simulation (Rust, Bevy) and a deterministic multi-agent LLM orchestrator (Rust). Key skills: Rust (tokio, actix-web, diesel, tonic gRPC, OpenAPI), C++20, CUDA, OpenCV, GStreamer, ROS 2, PX4/MAVLink, VIO / SLAM, MPPI, Docker, GitLab CI, TypeScript/React. English: C2.

Work Experience

Rust Developer (Backend) — SofIT Labs, Remote Oct 2023 – present

Pulsar — real-time photo/video enforcement system for road-traffic violations: multi-camera video ingest (GStreamer, industrial cameras), AI vehicle detection and tracking, licence-plate recognition, violation "checkers", evidence-material generation, REST/gRPC/MQTT integrations with regional back-offices. Rust workspace of 70 modules and 45 libraries running on NVIDIA Jetson (L4T) and x86 (CUDA / OpenVINO).

Responsibilities

- Design and implement backend features across the whole Pulsar stack: violation-detection logic, REST API (actix-web, OpenAPI), camera/peripheral drivers, external integrations, metrics.
- Investigate incidents from field complexes (traces, logs, recordings), reproduce them offline and ship hotfixes; maintain internal replay and debugging tooling.
- Drive engineering conventions (module contracts, route/handler conventions, test-portability rules) and the team's AI-assisted development workflow (agent instructions, orchestrated plan/implement/review loops).

Projects

- **Violation checkers** (Red Light, Stop Line, Speeding, Average Speed, Parking, Box Junction, Traffic Flow, Lane, Headlights) — owner of the checker code base: candidate-evidence model with automatic confirmation for Red Light / Stop Line, multiple stop lines and crossroad boundaries, priority between Average Speed and Speeding, report-point selector for Traffic Flow, track-merge fixes for opposite directions, taxi/bus whitelist, Uzbekistan red-light rules, PTZ / overview-camera survey frames for parking.
- **Security hardening for state certification (Uzbekistan)** — sole implementer: brute-force guard module with login throttling, session invalidation on logout, clickjacking and path-traversal protection, role-based route authorization audit, removal of Swagger from production builds.
- **Video collages and Track Frame Sampler** — horizontal / vertical video collages with parking-time compression robust to occlusion and frame jitter, quality/bitrate settings, generation speed-up; CLI tool that extracts annotation collages (3x3 / 4x4 / video) from recordings by trace timestamps.
- **External integrations** — gRPC "Doris Control" export v2 (tonic; photo/video upload, timeouts, opt-out), FVF MQTT monitoring service (Krasnodar), UrbanTech and Admin-Practice monitoring metrics, Prometheus metrics for video-archive and certification, public API consistency.

- **Hardware and cameras** — FFI bindings (bindgen) for the HIKROBOT camera SDK, Daheng camera flash backend and stream recovery after incomplete frames, traffic-light phase calibration and persistence, polarization-filter and floodlight fixes.
- **Platform work** — migration of web routes onto the internal typed Bus, OpenAPI 3.1 schema migration (apistos / schemars), unified pagination, DB migrations (Diesel), long-term frame cache for videos, removal of legacy slots and channels, unit-test coverage for the violations module, trace-runner replay tool.
- **collage-hub** (solo, 2026) — standalone Rust service (actix-web, reqwest) that proxies collage templates between a GitLab settings repository and Pulsar complexes: opaque-token sessions, CORS allow-list, YAML validation, demo rendering; 3.7K LOC, full API reference.
- **Web UI contributions** (2026) — violation-settings screens, video-archive recording status and streamed MP4/TS download, manual traffic-light phase calibration UI (React 19, TypeScript, Reatom, Mantine, Vite, Bun tests).

Results

- 100+ tasks delivered, 825 commits, 100K+ lines added in a 530K-LOC codebase over 3 years; contributed to 68 of 69 modules.
- 58 production hotfixes for field complexes; security package for certification delivered in under 2 weeks (5 tasks).
- Video-collage pipeline designed, implemented and tuned on field recordings in 1 week.

Technologies: Rust (tokio, actix-web, apistos/OpenAPI, diesel, tonic/prost gRPC, rumqttc MQTT, serde, tracing, metrics), OpenCV, GStreamer, CUDA / NVIDIA Jetson, FFI (bindgen, cxx), SQLite, Prometheus, Docker, GitLab CI, TypeScript / React.

Rust Developer — WiseTech, Remote Sep 2022 – Apr 2023

HFT trading system: development and refactoring of exchange connectors, added new crypto-exchanges, ported trading indicators from Python to Rust. **Technologies:** Rust, tokio, futures, WebSocket, REST, multithreading.

Rust Developer — Uddùg, Remote Nov 2021 – Sep 2022

Projects

- **index-daemon** (ICEX) — lead developer (283 of 323 commits): rewrote a C++ market-data daemon in Rust; real-time ticker / trades / order-book ingestion from 13 crypto-exchanges over WebSocket and REST, coin average price and volume calculation, historical storage in sled with candles, JSON-RPC WebSocket server with subscriptions, graceful shutdown, config via files or env, integration tests, Dockerfile and GitHub Actions CI.
- **Blockchain** — smart contracts and price oracles for Oasis, NEAR and Solana; Substrate pallet implementing ERC-20 semantics; Ethereum transaction crawler; ported a Go library to Rust.

Technologies: Rust, tokio, WebSocket, REST, sled, JSON-RPC, Substrate, NEAR, Oasis, Solana, Docker, GitHub Actions.

PHP Web Developer — FirstBit, Rostov-on-Don Oct 2020 – Dec 2021

Development and support of clients' websites.

PHP Web Developer — x5studio, Rostov-on-Don Aug 2018 – Sep 2020

Development and support of clients' websites.

Independent R&D Projects (2026, open source)

PX4 ROS 2 Drone Navigation — C++20, CUDA GitHub | Demo

Autonomous UAV navigation (PX4 quadrotor in offboard control) through GNSS-denied indoor, urban and underground spaces with no prior map; ROS 2 Jazzy, PX4 SITL and Gazebo Harmonic on the DARPA SubT Urban Circuit location. Localization by own stereo visual-inertial odometry (VIO, stereo MSCKF) or lidar-inertial odometry (LIO), fused into the PX4 EKF2 as external vision, with drift corrected by map-based relocalization and

point-to-plane ICP registration (SLAM in the broad sense); the camera profile flies with no GNSS, magnetometer or lidar. Benchmarked on real recordings against OpenVINS on EuRoC MAV: ATE 0.050, 0.046 and 0.115 m on V1_01_easy, V1_02_medium and V1_03_difficult (OpenVINS: 0.057, 0.080 and 0.058 m). Probabilistic 3D voxel occupancy mapping (OctoMap-style) from stereo depth or 3D lidar with incremental GPU ESDF; persistent incremental D* Lite 3D path planner on a sparse lattice with free-space topology for tunnels and shafts; GPU MPPI local planner (model predictive path integral, sampling-based MPC) for obstacle and collision avoidance under a braking contract tied to measured sensor range; return to home (RTH) when a goal is proven unreachable; cooperative multi-drone air traffic with intent exchange. Deterministic whole-stack replay of recorded ROS 2 bags as a regression gate. 188K LOC of C++20 / CUDA plus 35K LOC of Python tooling, 1,550+ unit tests, 2,580+ commits, releases up to v0.7.0 (Jun–Oct 2026), 168 GitHub stars and 33 forks, Dockerized NVIDIA dev environment.

Drone Swarm Lab — Rust GitHub

Mission-level coordination runtime for autonomous drone fleets: deterministic simulation, distributed CBBA task allocation and strategy comparison benchmarks, heartbeat-based membership and failure recovery with task reallocation, mission DSL and command IR, replay logs, safety validation layer, MAVLink transport with a PX4 SITL/SIH supervisor for multi-agent mission upload and execution. 12 crates, 90K LOC, 1,195 tests, property-based testing (proptest), 540+ commits.

Multi-Agent Orchestrator — Rust GitHub

Deterministic FSM-driven orchestrator for delegated coding-agent workflows: an executor agent (Claude / Codex / OpenCode) and an independent reviewer agent loop through investigate / plan / implement stages via a file-based transport until a quality gate is reached; hang detection, retries, Git preflight, batch task runner. Used daily to deliver production work on Pulsar. 12.8K LOC, 260+ commits.

Swarm — 2D self-organizing spacing network — Rust, Bevy

Agent simulation where agents self-organize into a spacing network from local rules only (repulsion / attraction, wall repulsion, escape mode, population events); pure simulation kernel with property tests plus a Bevy real-time visualisation.

NEAR Messenger — Rust, near-sdk (2024) Contract | Client

On-chain messenger prototype: NEAR smart contract (WASM) and a Rust client.

Education

2013 – 2017 B.Sc. in Computer Science and Programming, **Taganrog State Radio Technical University** (Southern Federal University), Taganrog, Russia

Tech Skills

Languages	Rust (main, 5 years commercial), C++20, CUDA, TypeScript, Python (tooling, 35K LOC: simulation, validation and evaluation scripts), PHP (3 years commercial), SQL
Rust ecosystem	tokio, actix-web, futures, serde, diesel, tonic / prost, apistos / schemars (OpenAPI), reqwest, tracing, metrics, clap, proptest, bindgen / cxx FFI, near-sdk, Substrate
Computer vision & media	OpenCV, GStreamer, NVIDIA Jetson / CUDA, OpenVINO, industrial cameras (HIKROBOT, Daheng), RTSP
Robotics & simulation	UAV autonomy in GNSS-denied environments, VIO (MSCKF), LIO, SLAM / relocalization, EKF (PX4 EKF2 fusion), ICP, 3D occupancy mapping (OctoMap-style), ESDF, path planning (D* Lite, lattice), MPPI / MPC, obstacle avoidance, ROS 2, PX4 (SITL / SIH, offboard), Gazebo, MAVLink, Bevy
Infrastructure	Docker, GitLab CI, GitHub Actions, Prometheus, MQTT, gRPC, WebSocket, JSON-RPC, SQLite, sled, Linux
Practices	message-bus modular architecture, property-based and integration testing, deterministic replay and replay-based debugging, AI-assisted engineering (agent orchestration, plan / implement / review loops)
Languages spoken	English (C2), Russian (native)